Whole Wheat Garlic Breadstick

Bread/Grain

Iowa Gold Star Recipe

Recipe Source: Healthier Kansas Recipe 107

Ingredients	100 Servings		____Servings		Directions
	Weight	Measure	Weight	Measure	
White whole wheat flour	1 lbs 14 oz				1. Place flour, dry milk, sugar, yeast and salt in mixer bowl. 2. Blend with dough hook for approximately 2 minutes on low speed. 3. Add oil and blend for approximately 2 minutes on low speed. 4. Add first amount of water to the dry ingredients. If dough appears too stiff, add up to ½ cup extra water per 100 servings. More or less water can be added as needed. 5. Knead dough on medium speed for 8-10 minutes until dough is properly developed. Use the gluten stretch test. (At 8 minutes, stretch the dough like a rope. If it breaks, continue to mix 2 additional minutes). 6. Form dough into 1 pound balls and spray with non-stick cooking spray or cover with plastic wrap. Once all dough balls have been formed, begin shaping. 7. Form rolls from dough by pinching off 1 oz pieces and shaping into sticks 4-5 inches in length. Place sticks in rows of 6 across and 8 down on sheet pans (18” x 26” x ½”) which have been lightly coated with non-stick cooking spray. 8. Place in a warm area (about 90°F) until double in size, approximately 45-60 minutes. 9. Combine butter, olive oil and granulated garlic in small bowl. Brush mixture on top of breadsticks. 10. Bake until lightly browned: Conventional oven: 400°F for 18-20 minutes Convection oven: 350°F for 12-14 minutes
All-purpose flour, enriched*	1 lbs. 12 oz				
Dry milk, instant, non-fat	4 oz	1 ¾ cup			
Sugar, granulated	6 oz	1 ¼ cup 3 Tbsp			
Yeast, instant, dry	2 oz	½ cup			
Salt	1 ¾ oz	1 Tbsp 1 tsp			
Vegetable oil*	7 oz	1 2/3 cup (+ up to 1 cup if dough is stiff)			
Water (calculate water temperature by subtracting the temperature of the dry ingredients from 145°F)		1 qt 2 Tbsp			
Non-stick cooking spray		As needed			
Butter, melted		1/2 cup			
Olive oil		1/2 cup			
Granulated garlic		1 Tbsp			

					Turn rolls half-way through baking time to promote even baking. Recommended internal temperature for baked whole wheat rolls is 196-198°F. Notes: • Dough made with whole flour will absorb more water and requires increased rising/proofing time. Mixing time is less because the bran from the whole grain cuts through developing gluten strands with increased mixing resulting in a product with low volume. Even an additional one minute mixing time with whole grains can make a difference in quality. Figuring water temperature so that the final dough temperature is ~80°F is essential for quality bread products.

*USDA Foods

Serving Size	1 Serving Provides	Yield
1 oz roll	1 oz eq grains.	

Nutrients Per Serving

Calories	100	Sodium	110 mg	Vitamin A	33 IU
Total Fat	4 g	Carbohydrates	15 g	Vitamin C	0.06 mg
Saturated Fat	1 g	Dietary Fiber	1.4 g	Calcium	17 mg
Trans Fat	0 g	Protein	3 g	Iron	0.72 mg
Cholesterol	3 mg	Calories from Total Fat	36 %	Calories from Saturated Fat	9 %